

MOHAMMAD SAMIR AHMAD AL QARAGHULI

5D BIM & Digital Construction Specialist | Cost Manager | BIM-QS Integration

Dubai, UAE | +971 56 396 1932 | mohammad7samir.ahmad@gmail.com

[linkedin.com/in/mohammad-ahmad-4518391a0](https://www.linkedin.com/in/mohammad-ahmad-4518391a0)



PROFESSIONAL SUMMARY

Civil Engineer and 5D BIM / cost-management specialist with 6+ years of experience across aviation, infrastructure, residential, mixed-use and digital-product environments in the UAE, Saudi Arabia, Malaysia and Iraq. Combines pre- and post-contract quantity surveying with Revit-to-CostX workflows, model-based quantification, BIM data classification, BOQ automation, model QA/QC, Autodesk Construction Cloud, Power BI and software product development. Experienced in NRM, POMI, CESMM4 and ISO 19650-aware delivery, translating design information into controlled quantities, cost plans, tender BOQs, variation assessments, commercial reports and decision-ready data. Founder of Threshold Technologies and developer of construction-technology products focused on BIM commercial readiness, field reporting, cost intelligence and technical learning.

TARGET ROLE AND SPECIALIST STRENGTHS

Target roles: BIM Manager / BIM Manager - Digital Delivery | 5D BIM Specialist / BIM-QS Integration Specialist

Career direction: Pursuing project- or site-based BIM management and 5D BIM opportunities where model governance, multidisciplinary coordination and information management are integrated with quantities, cost and commercial decision-making. Recent consultancy, R&D and product-development work has been deliberately focused on this specialist area.

- Integrated BIM-commercial background: combines Civil Engineering, quantity surveying and cost management with Revit, Navisworks, Autodesk Construction Cloud and CostX.
- 5D BIM delivery: Revit parameter strategy, NRM/POMI/CESMM4 classification, model-to-BOQ mapping, quantity validation, CostX-ready information and design-change/cost reconciliation.
- BIM management and coordination: ISO 19650-aware workflows, model QA/QC, federated-model review, clash detection, controlled CDE submissions, multidisciplinary coordination and stakeholder reporting.
- Automation and digital innovation: C#, WPF, Dynamo, Power BI and structured-data workflows, including a 132-check Commercial Readiness Standard and tools for measurable, auditable model information.

TECHNICAL COMPETENCIES

5D BIM and Digital Delivery: BIM-to-BOQ workflows; Revit parameter strategy; model classification and mapping; model-based quantity take-off; federated-model review; IFC workflows; LOD/data-readiness checks; clash-informed cost impact; controlled model revisions; CDE workflows; EIR/BEP and ISO 19650 awareness

Cost Management: cost planning; estimating; Bills of Quantities; pre-tender estimates; tender documentation and analysis; value engineering; benchmarking; interim valuations; payment certification; variations and change control; claims review; cost reporting; forecasting; final accounts

Measurement Standards: RICS NRM 1/2/3; POMI; CESMM4; measurement rules; structured work breakdowns; model-to-classification mapping; commercial data validation

Data and Automation: Power BI dashboards; Excel and VBA; C# Revit add-in development; Dynamo; automated classification; BOQ mapping; structured CSV/data export; design-change and quantity reconciliation

Coordination and Delivery: Navisworks clash detection; clash matrices; ACC/BIM 360 document control; multidisciplinary coordination; design review; QA/QC; technical submittals; as-built review; stakeholder workshops and client reporting

SOFTWARE AND TECHNOLOGY

BIM / Design: Autodesk Revit, Navisworks Manage, AutoCAD, Civil 3D, BIM 360, Autodesk Construction Cloud, Twinmotion, Dynamo, IFC, RVT, NWD, DWFX

Cost / Commercial: CostX, WinQS, DimX, Cubicost, Microsoft Excel, Microsoft 365

Data / Development: Power BI, Excel VBA, C#, WPF, Next.js, TypeScript, Supabase, GitHub, Netlify, AI-assisted product development

Structural: Prokon, STAAD.Pro, Autodesk Robot Structural Analysis

PROFESSIONAL EXPERIENCE

Founder and 5D BIM / Digital Construction Lead | Threshold Technologies FZE LLC

Jun 2026 - Present | Ajman / Dubai, UAE

- Founded and lead a UAE technology company developing BIM, cost-management, field-reporting, education and AI-enabled engineering products.
- Define product architecture and technical workflows connecting Revit, CostX, ACC, Power BI and structured construction data.
- Develop ISO 19650-aware information-management, model-governance and commercial-readiness frameworks for consulting and software use cases.
- Lead product validation, client demonstrations, technical documentation, developer coordination and commercial positioning across Threshold Technologies, THS Studio and Modern Engineering Lab initiatives.

Independent 5D BIM and Construction Technology Consultant | Independent Practice

Jan 2026 - Present | Dubai, UAE

- Design model-to-BOQ workflows for architectural, structural and MEP models, including classification, parameter mapping, quantity validation and CostX-ready outputs.
- Develop proof-of-concept tools and dashboards for BIM commercial readiness, BOQ benchmarking, design-change tracking, cost reporting and digital delivery.
- Produced an NDA-protected structural test-model workflow for Omnium, including automated NRM classification, BOQ description mapping, CostX quantification and DWFX export preparation.

Quantity Surveyor / Cost Consultant | Blair Anderson Limited

Jul 2025 - Jan 2026 | Dubai, UAE

- Prepared full Bills of Quantities, cost plans, tender documents and detailed measurements for building, infrastructure, landscape, structural and MEP packages using CostX, WinQS and DimX.
- Measured and classified complex model/drawing packages, including ten MEP models, raft and reinforcement, post-tensioned slabs and beams, security buildings, cash-office works and landscape ITS/SCADA systems.
- Supported tender analysis, interim valuations, variation measurement, cost reporting and client/consultant coordination.
- Applied NRM and POMI measurement structures and developed BIM-enabled workflows to improve quantity consistency and BOQ traceability.

Project Surveyor / Cost Manager - BIM-Enabled Delivery | AECOM Middle East

Jun 2022 - Mar 2025 | Dubai, UAE

- Delivered cost estimates, cost plans, pre-tender estimates and BOQs across aviation, infrastructure, residential, hospitality, landscape and mixed-use projects using CostX, Revit and AutoCAD.
- Performed model-based measurement, mapped federated Revit models into CostX and reconciled design revisions, quantities and tender BOQs.

- Reviewed interim payment applications, payment certificates, contractor claims and final-account information; assessed variations and change orders with supporting cost breakdowns and commercial impact analysis.
- Prepared tender analyses, bidder comparisons and tender recommendation inputs for procurement and client decisions.
- Built Power BI dashboards for estimates, benchmarks, design changes, model revisions, bidder comparisons, cost risks and commercial exposure.
- Managed multidisciplinary model submissions through BIM 360, supported controlled review cycles, performed Navisworks clash detection and prepared clash matrices and visual coordination reports.
- Worked with design, BIM, commercial, contractor and client teams to align model information with measurement rules, cost structures and deliverable requirements.

Quantity Surveyor | Fan Al Ebtikar Architectural Engineering Consultancy

Dec 2021 - Jun 2022 | Dubai, UAE

- Prepared ROM estimates and early cost-planning inputs for a group of five residential villas.
- Supported tender evaluation, scope-change reviews and contractor negotiations; conducted site quantity verification and progress inspections.
- Reviewed technical submittals and as-built drawings and coordinated technical/commercial queries with contractors and consultants.
- Supported site QA/QC and compliance with approved drawings, specifications and authority requirements.

Design / CAD and BIM Technician (Remote) | Al Etihad Engineering Consultancy

2020 - 2022 | Baghdad, Iraq

- Produced 2D architectural drawings in AutoCAD and developed coordinated Revit models and LOD 200 architectural, structural and interior-design documentation.
- Delivered more than 20 villa designs plus mosque, school, hotel and sports-centre assignments; produced client visualisations and walkthroughs in Twinmotion.
- Supported structural element design and analysis using Prokon, STAAD.Pro and Autodesk Robot Structural Analysis.

Site Engineer | Cityscape Global Engineering Consultants

Apr 2020 - Dec 2020 | Kuala Lumpur, Malaysia

- Supported quantity take-offs, cost breakdowns and weekly progress reporting for a high-school project and a smart interlocking-brick residential unit.
- Measured concrete, pavement, curb and site-development works and contributed procurement and cost-planning inputs.
- Assisted preliminary highway-alignment and cost-impact studies and an urban traffic-congestion case study using design data and scenario modelling.

CONSTRUCTION TECHNOLOGY PRODUCTS AND R&D

BIM Commercial Readiness and 5D Classification Platform | Product

- Developed the Commercial Readiness Standard (CRS) v1.1 with 132 structured checks across six control groups for model data, geometry, classification, measurement and delivery readiness.
- Designed BIM-to-BOQ workflows that classify Revit data to NRM, POMI and CESMM4, generate CostX-ready grouped models and derive item descriptions from model specifications and parameters.
- Directed product UI, developer handover, technical documentation, learning content, demonstrations and go-to-market preparation.

DiSam Digital Ultimate Tool - Revit Add-in | C# / WPF Product Development

- Designed Revit functions for automated NRM Level 1-3 classification, POMI/WBS coding, BOQ mapping, model isolation/review, structured CSV export and commercial-readiness scoring.
- Defined parameters including Class_NRM_Level1/2/3, POMI_Code, WBS_Code, Rate, Depth and Width to support measurable and auditable model data.

Site Report Engine | Product

- Designed a bilingual construction field-reporting workflow covering project, visit, observation, issue, instruction, response, verification, close-out and immutable PDF snapshots.

- Specified role permissions, lifecycle controls, photo captions, attachment handling, deduplication and issuance safeguards using Next.js, Supabase and automated PDF generation.

SELECTED PROJECT PORTFOLIO

Dubai International Airport - Terminal 2 Expansion | AECOM

- Delivered design-stage measurement, CostX BOQs, scope validation, estimate reconciliation, value engineering and a Power BI cost/change dashboard.

Dubai International Airport and Dubai Airshow Packages | AECOM

- Supported Dubai Airshow 2023 infrastructure expansion; proto-cabin offices and Gate 5; exhibition-hall toilets; PRA assets and battery chargers; electrical utilities and pumping-station works; Concourse C modifications; and roof-joint repairs.
- Led detailed measurement of Concourse C roof-sealant replacement and supported cost management through tender documentation and estimate validation.

The Address Residences Dubai Opera | AECOM | 850+ serviced apartments

- Managed fit-out and MEP variation assessments, reviewed subcontractor claims and as-built information, and supported commercial negotiations.

NEOM - Sindalah Island | AECOM

- Revalidated BOQs and assessed more than 75 variations / work-order changes across marine, commercial, retail, residential and infrastructure packages, coordinating design and commercial information.

NEOM - The Line Landscape | AECOM

- Prepared infrastructure pre-tender estimates across development zones, aligned design scope with cost plans and supported compilation of the overall PTE.

NEOM Bay Airport / NEOM International Airport | AECOM

- Supported design-stage cost planning, reconciliation and value engineering and contributed to final design cost reporting for approval.

Al Haffar / Aldar Large-Scale Residential Development | AECOM | 4,000+ villas

- Built a CostX model-mapping workflow across ten villa designs and three finish options, generated model-based BOQs and supported value engineering with Power BI insights.

Emaar High-Rise Residential Projects | AECOM

- Measured and evaluated changes for luxury residential towers, including Il Primo A4, provisional quantities, fit-out items and sanitary fixtures; supported contractor-submission reviews and approvals.

North Riyadh Development | AECOM | Majid Al Futtaim

- Priced roadworks, utilities and landscape packages, performed bidder analysis and contributed to formal tender evaluation reporting.

Jeddah North Infrastructure and Park Facilities | AECOM

- Prepared Navisworks clash matrices and Power BI reports and supported Civil 3D model updates and interdisciplinary issue resolution.

House of Performing Arts | AECOM | Confidential

- Updated cost plans through design stages, measured design changes, maintained risk and reporting inputs and attended weekly design/progress reviews.

AMIA Airport / DAEP | AECOM

- Performed BIM-based quantification for 50+ km of automated people-mover systems and maintenance facilities and developed Power BI estimate, benchmark and bidder-comparison dashboards.

Blair Anderson Building and Infrastructure Assignments | Blair Anderson

- Prepared POMI-based security-building BOQs; measured ESO and Cash Office Phase 2 works; classified ten MEP models; and measured raft, reinforcement, post-tensioned slabs/beams and landscape ITS/SCADA packages.

EDUCATION

Bachelor of Civil Engineering (Hons): Limkokwing University of Creative Technology, Kuala Lumpur, Malaysia | 2017 - 2021

Foundation in Science: UCSI University, Kuala Lumpur, Malaysia | 2016 - 2017

PROFESSIONAL DEVELOPMENT AND KNOWLEDGE SHARING

- Delivered internal Microsoft Power BI teaching support through AECOM University.
- Continuous professional development in 5D BIM, digital construction, AI-assisted engineering, RICS measurement standards, ISO 19650, cost automation and construction technology product development.

LANGUAGES

Arabic: Native

English: Professional working proficiency